Is the Bayesian OOD Advantage Just Softness? Optimizer Time, Mean-Field VI and the Exact Posterior

Anonymous authors
Paper under double-blind review

Abstract

Bayesian neural networks are often reported to detect out-of-distribution (OOD) inputs better than deterministic networks, but their predictions are also softer everywhere, and softness can be bought with a temperature. We compare each method with a deterministic network whose temperature is fitted, on validation data only, to the method's mean in-distribution entropy. In pre-registered experiments with small MLPs on MNIST-family data, mean-field Bayes by Backprop (BBB) is no better than this control at standard budgets on MNIST, but with more optimizer steps it beats it by 0.06–0.08 AUROC, also against step-matched networks and ensembles, weight noise, and the best temperature chosen with the OOD set. We derive why: under Adam the KL gradient on the variance parameters is nearly constant, so posterior standard deviations grow on a learning-rate $\times$ steps clock almost independent of the data size; under SGD they do not, and no advantage appears. Under a prior where full-batch HMC converges, the HMC posterior also beats the control, more than a MAP ensemble under the same prior, and BBB's mutual information matches its advantage on KMNIST, though by a structurally different solution. The advantage is a far-OOD effect: on held-out MNIST classes it disappears.

1 Introduction

A standard classifier gives confident predictions on inputs unlike its training data.[1] Bayesian neural networks average predictions over a distribution of weights (Blundell et al., 2015), which should raise their uncertainty on unfamiliar inputs, and the usual evidence is that their predictive entropy separates in-distribution from out-of-distribution (OOD) inputs better than a deterministic network's.

That evidence has a confound. An averaged prediction is softer than a single network's on *every* input, and temperature scaling (Guo et al., 2017) softens a deterministic network at no cost. Temperature is a known ingredient of OOD detection (Liang et al., 2018), uncertainty benchmarks include temperature-scaled baselines (Ovadia et al., 2019), and Ashukha et al. (2020) argue that uncertainty methods should be compared after temperature scaling. Whether Bayesian posteriors are intrinsically good OOD detectors has also been questioned (Henning et al., 2021). We make the confound the centre of the comparison: every method is compared with a deterministic network softened, on validation data only, to that method's own in-distribution entropy, and we ask whether the method ranks OOD inputs as more uncertain than in-distribution inputs better than a network of the same softness.

We study mean-field Bayes by Backprop (BBB), the canonical variational BNN, on small MLPs where every question can be asked directly, including how the answer compares with the exact posterior. Every experiment had its hypotheses, predictions and falsification criteria written down before it ran; Appendix H lists all 87 predictions with their outcomes, 31 of which were falsified. Our findings are:

[1] Use of AI tools: an LLM-based assistant (Claude, Anthropic) was used extensively throughout this work: to suggest experiments, analyses and parts of the derivation in §4, to draft the pre-registered protocols, to implement, run and debug the experiment and analysis code, to write and edit the text, and, as a separate agent, to audit code and claims (Appendix G). The authors directed the study, decided what to run and report, and take full responsibility for all content.

1. **The advantage follows optimizer steps.** On MNIST at standard budgets BBB is no better than the softness-matched network. With more optimizer steps it beats it by +0.06 to +0.08 AUROC at every training-set size we tried, also against step-matched plain networks and deep ensembles, random weight noise, a plain network at the best temperature chosen with knowledge of the OOD set, and on correctly classified images only. With Fashion-MNIST as the in-distribution data the long-training advantage replicates (+0.08 to +0.21), but there it is already present at the standard budget and a step-matched ensemble ties BBB on one OOD set. On a near-OOD split (trained on digits 0–4, tested against digits 5–9) the advantage largely disappears (+0.011, no better than step-matched deterministic models), so our claims concern far-OOD detection (§3).

2. **An optimizer clock explains when (on MNIST).** Under Adam the gradient of the KL term on each variance parameter is nearly constant, so the posterior standard deviations grow by about one learning rate per step, independently of the data size. This predicts, with no fitted constants, the $1/\eta$ timing under Adam, its independence of N, and the N-dependent growth under SGD, where BBB then shows no advantage. IVON, a natural-gradient method whose variances scale with N, showed none either in our configuration (§4).

3. **The HMC posterior has the advantage too.** Under a prior of scale 0.3, where full-batch HMC converges (8 chains, three data subsets per n), the posterior beats the softness-matched network by +0.05 to +0.11. BBB's mutual information on KMNIST matches it within 0.012; on Fashion-MNIST and with the entropy score BBB recovers only part of it. Under the default $\mathcal{N}(0, 1)$ prior HMC does not mix at this scale (§5).

4. **BBB gets there by a different route.** Each weight's marginal under the exact posterior is the prior, and sampling weights independently from those marginals gives chance accuracy, as expected for a model with far more parameters than training points. Long-trained BBB is instead overpruned: a few precise means carry the prediction. A first-layer noise-to-signal ratio computed from its means and variances matches its sampled far-OOD detection within 0.02 AUROC, but on far-OOD sets a trivial pixel statistic tracks BBB's per-image uncertainty as well, so this is a cheap score rather than an explanation (§6).

A caveat applies throughout: the standard far-OOD pairs used here are nearly solved by a model-free rule that flags ink on pixels that are always background in the training data (AUROC 0.967–1.000). The models do not rely on it (§3), but it means these benchmarks test a narrow kind of OOD detection. On a held-out-class split that the rule cannot solve (AUROC 0.57), BBB's advantage largely disappears (§3).

What the paper offers is therefore mainly methodological: a control that separates a Bayesian method's OOD signal from its softness, evidence that for mean-field VI that signal depends on the training run rather than the method, and an account of when it appears. It is not a new OOD detector, and the effect it studies is confined to far-OOD inputs.

The practical consequence is that a reported OOD gain of a variational BNN is a property of the training run as much as of the method: comparisons should report the optimizer, learning rate and budget, the uncertainty score, and a softness-matched baseline.

2 Setup

Data. In-distribution (ID) data are class-stratified training subsets of MNIST of size $n \in \{500, 2{,}000, 10{,}000\}$ (one subset per seed), a fixed validation pool of 5,000 further training images, disjoint from every subset, and the MNIST test set. OOD sets are the Fashion-MNIST (Xiao et al., 2017) and KMNIST (Clanuwat et al., 2018) test sets, normalised with the ID statistics. In Table 4 the roles are swapped: Fashion-MNIST is ID, MNIST and KMNIST are OOD; and in a held-out-class split, the network is trained on digits 0–4 and the test digits 5–9 are a near-OOD set.

Models. All models use a 784–256–256–10 ReLU MLP (width 100 for the HMC comparison). *DNN*: Adam (10^{-3}, cosine schedule), 40 epochs, batch 64. *Deep ensemble* (Lakshminarayanan et al., 2017): 5 DNNs. *MC*

dropout (Gal & Ghahramani, 2016): $p = 0.1$, 50 samples. *Last-layer Laplace* (Daxberger et al., 2021): full generalised Gauss–Newton covariance over the last layer, prior precision tuned on validation NLL. *IVON* (Shen et al., 2024): the official implementation with effective sample size N and the same $\mathcal{N}(0, 1)$ prior as BBB. *BBB* (Blundell et al., 2015): mean-field Gaussian posterior, $\mathcal{N}(0, \sigma_p^2)$ prior with $\sigma_p = 1$ unless stated, and per-example negative ELBO

$$\mathcal{L} = \frac{1}{B} \sum_{i \in \text{batch}} -\log p(y_i \mid x_i, w) + \frac{1}{N} \mathrm{KL}\big[q(w) \,\|\, p(w)\big], \qquad w = \mu + \sigma \odot \varepsilon, \tag{1}$$

with the KL weight warmed up linearly, $\sigma = \mathrm{softplus}(\rho)$, $\rho_0 = -5$ ($\sigma_0 \approx 0.007$), Adam (10^{-3}), gradient clipping and 50 posterior samples at test time. The standard budget is 150 epochs at $n \geq 2{,}000$ (4,800 steps at $n = 2{,}000$, 23,550 at $n = 10{,}000$) and the same 4,800 steps at $n = 500$ (600 epochs). Dividing the KL term by the number of mini-batches instead of N, an easy slip, over-weights it by about the batch size and costs 3 points of accuracy (Appendix A).

The softness-matched control. For a method M we fit a temperature T to a DNN's logits so that its mean predictive entropy on the validation set equals M's; no test or OOD data enter the fit. We report

$$\Delta = \mathrm{AUROC}(M) - \mathrm{AUROC}(\mathrm{DNN}_T), \tag{2}$$

where AUROC measures the separation of test from OOD images by predictive entropy; $\Delta > 0$ means M ranks OOD inputs better than a network of the same mean softness; the shape of the entropy distribution is not matched. The MI-based Δ uses M's mutual information (Depeweg et al., 2018) instead, against the same softened network's entropy. A single network has no mutual information, so this compares two different scores: it asks whether M's epistemic score beats the best softness-based score of a plain network. It is only meaningful for a model that has learned the task, since a model whose posterior samples disagree on every input also has a large MI-based Δ (§4). Comparisons are paired by seed with bootstrap confidence intervals and sign counts; Wilcoxon tests are reported only with 10 seeds.

Pre-registration. Each experiment's protocol (hypothesis, predictions, falsification criterion, configuration) was written before it ran: committed to version control for the first nine experiments and, for the rest, with a SHA-256 hash of the protocol recorded before launch. These records are self-attested rather than independently timestamped. Experiments are referred to by their identifiers (E1–E23); Appendix H lists every prediction.

3 The advantage follows optimizer steps

At standard budgets, BBB matches the control. Table 1 compares each method with its own softness-matched DNN over 10 seeds (E2). At $n = 500$ and $n = 2{,}000$, BBB is *worse* than the control (0/10 seeds better on KMNIST at both, 0/10 and 3/10 on Fashion-MNIST). At $n = 10{,}000$ it is better in all 10 seeds. Deep ensembles beat the control by +0.01 to +0.02 at every n, MC dropout roughly ties it, and last-layer Laplace is below it.

The jump at $n = 10{,}000$ is about steps, not data. With a fixed number of epochs, more data also means more optimizer steps (23,550 at $n = 10{,}000$ against 4,800). Decoupling the two (E4, Figure 1): with the same data, 23,550 steps give $\Delta = +0.06$ to $+0.08$ in 5/5 seeds at every n, including $n = 500$, while 4,800 steps give a negligible one at most, including at $n = 10{,}000$ (-0.006 on Fashion-MNIST, $+0.008$ on KMNIST).

Stronger controls do not close the gap. Long-trained BBB (23,550 steps) also beats controls that share its training budget or its noise (E12, Table 2; 5 seeds, $n \in \{2{,}000, 10{,}000\}$). A plain network and a 5-member ensemble trained for the same number of steps stay below it in all 5 seeds in every setting. Adding Gaussian noise to every layer of a trained plain network, scaled per layer and matched to BBB's validation entropy, gives AUROC 0.81–0.89, and BBB beats it by +0.10 to +0.15; even the best noise scale chosen

Table 1: Each method against a DNN softened to its own validation entropy (entropy-AUROC difference Δ, mean and 95% bootstrap CI over 10 seeds; standard budgets; E2).

n	method	Δ Fashion-MNIST	Δ KMNIST
500	BBB	−0.046 [−0.063, −0.032]	−0.035 [−0.041, −0.028]
500	ensemble	+0.014 [0.003, 0.027]	+0.010 [0.008, 0.012]
500	MC dropout	−0.003 [−0.016, 0.009]	−0.019 [−0.021, −0.018]
500	last-layer Laplace	−0.072 [−0.088, −0.055]	−0.112 [−0.122, −0.102]
2,000	BBB	−0.024 [−0.048, −0.001]	−0.010 [−0.014, −0.007]
2,000	ensemble	+0.015 [0.007, 0.022]	+0.015 [0.013, 0.018]
2,000	MC dropout	+0.007 [0.002, 0.012]	−0.010 [−0.012, −0.007]
2,000	last-layer Laplace	−0.040 [−0.047, −0.032]	−0.043 [−0.047, −0.038]
10,000	BBB	**+0.048** [0.038, 0.059]	**+0.069** [0.065, 0.074]
10,000	ensemble	+0.016 [0.008, 0.026]	+0.023 [0.021, 0.025]
10,000	MC dropout	−0.010 [−0.018, −0.001]	−0.007 [−0.010, −0.003]
10,000	last-layer Laplace	−0.014 [−0.023, −0.006]	−0.012 [−0.016, −0.008]

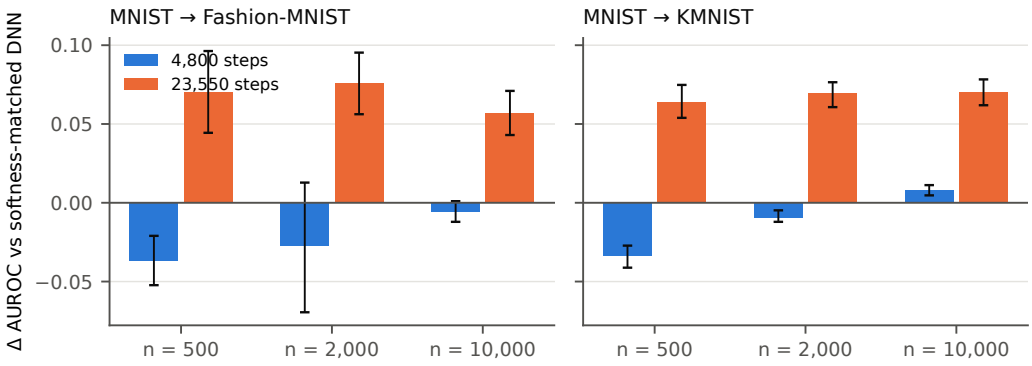


Figure 1: BBB against its softness-matched DNN with the same training data and two step budgets (5 seeds, 95% CIs; E4).

with knowledge of the OOD set stays +0.07 to +0.10 below BBB. So the advantage is not produced by *any* weight noise. Among post-hoc deterministic detectors, a kNN detector on normalised penultimate features comes closest (within +0.006 to +0.023, and tying BBB at $n = 10{,}000$ on Fashion-MNIST, 2/5 seeds better).

Table 2: Long-trained BBB (23,550 steps) minus each control (entropy AUROC; range over $n \in \{2{,}000, 10{,}000\}$ and, for feature detectors, two base networks; 5 seeds each; E12).

control	Fashion-MNIST	KMNIST
plain network + weight noise (softness-matched)	+0.11 to +0.14	+0.10 to +0.15
step-matched plain network, softened	+0.040 to +0.087	+0.075 to +0.078
step-matched 5-member ensemble	+0.041 to +0.058	+0.058 to +0.075
Mahalanobis on features	+0.021 to +0.109	+0.030 to +0.059
kNN on features	+0.006 to +0.023	+0.017 to +0.023

Softness matters, but does not explain the advantage. Is the softness-matched control a strawman? A post-hoc sweep over temperatures (exploratory; Table 3) shows that it is not: temperature moves the plain network's entropy AUROC by about 0.10 between the worst and best temperature on a grid from 0.25 to 50, and the best temperature, chosen *with* knowledge of the OOD set, gains +0.035 over $T = 1$ on average. But long-trained BBB beats even this oracle-tempered network in every seed of every setting (50/50; by

+0.042 to +0.070 with MNIST as ID and +0.079 to +0.196 with Fashion-MNIST). Nor is the advantage a side effect of BBB being uncertain on misclassified in-distribution images: restricted to test images that both BBB and the plain network classify correctly, Δ stays positive in every setting (+0.046 to +0.062 on MNIST, +0.053 to +0.175 on Fashion-MNIST). With the cosine schedule used here, long-trained BBB on MNIST is as accurate as the plain network (0.933 against 0.925 at $n = 2{,}000$); on Fashion-MNIST it is less accurate (0.762 against 0.827).

Table 3: How much can temperature buy? Entropy AUROC of long-trained BBB and of the plain 40-epoch network at $T = 1$, at the softness-matched T, and at the best T on a log grid from 0.25 to 50 chosen with the OOD set (oracle); last column: Δ on test images both classify correctly. Means over 5 seeds (post-hoc, exploratory).

ID	n	OOD	BBB	DNN, $T = 1$	DNN, matched T	DNN, oracle T	Δ (both correct)
MNIST	500	Fashion	0.907	0.813	0.837	0.860	+0.058
MNIST	500	KMNIST	0.955	0.852	0.891	0.913	+0.046
MNIST	2,000	Fashion	0.951	0.862	0.875	0.888	+0.062
MNIST	2,000	KMNIST	0.968	0.871	0.899	0.909	+0.055
MNIST	10,000	Fashion	0.973	0.904	0.916	0.917	+0.050
MNIST	10,000	KMNIST	0.987	0.903	0.917	0.918	+0.058
Fashion	2,000	MNIST	0.980	0.779	0.844	0.845	+0.082
Fashion	2,000	KMNIST	0.827	0.716	0.745	0.747	+0.053
Fashion	10,000	MNIST	0.956	0.756	0.796	0.797	+0.117
Fashion	10,000	KMNIST	0.885	0.682	0.678	0.689	+0.175

Not a pixel shortcut. MNIST subsets have many always-background pixels, and a model-free rule that flags ink on such pixels separates MNIST from Fashion-MNIST and KMNIST almost perfectly (AUROC 0.967–1.000), better than any model here, a real weakness of these benchmarks. The models do not rely on it: resetting those pixels to background in the OOD images leaves every method's Δ essentially unchanged (E5, Appendix B). The advantage also needs posterior noise in both hidden layers together; noise in any single layer keeps essentially none of it (E6b, Appendix C).

With Fashion-MNIST as the in-distribution data. To test whether these results are specific to MNIST, we swapped the roles (E21; Fashion-MNIST as ID, MNIST and KMNIST as OOD; 5 seeds; Table 4). The long-training result replicates and is larger: BBB trained for 23,550 steps beats the softness-matched network by +0.136 (MNIST) and +0.082 (KMNIST) at $n = 2{,}000$, and by +0.160 and +0.206 at $n = 10{,}000$, in 5/5 seeds, and it beats the step-matched plain network in 5/5 seeds on both OOD sets. Three predictions failed or held only in part. First, BBB already beats the control at the standard budget (+0.080 and +0.066), so the absence of an advantage at standard budgets in Table 1 is specific to MNIST; one reading is that the clock of §4 fixes *when* the variances grow, not how much variance a dataset needs before the advantage appears. Second, the step-matched ensemble ties BBB on KMNIST (Δ +0.084 against +0.082; BBB higher in 2/5 seeds), although BBB's absolute entropy AUROC is higher (0.827 against 0.792). Third, long-trained BBB's MI-based Δ on MNIST is only +0.015. Long training also costs 6.5 points of accuracy here (0.762 against 0.827), and plain networks are much weaker detectors with Fashion-MNIST as ID (entropy AUROC 0.68–0.84), so even step-matched deterministic training gains a little (+0.018 to +0.040).

Held-out classes: the advantage is a far-OOD effect. A pre-registered held-out-class split (E22; 5 seeds) trains on MNIST digits 0–4 ($n = 2{,}000$) and treats the test digits 5–9 as OOD, with Fashion-MNIST and KMNIST for reference. Here the pixel rule is close to chance (AUROC 0.569, against 1.000 and 0.999 on the far-OOD sets), so a detector must use more than where the ink is. The plain network reaches 0.890 on held-out digits. Long-trained BBB's advantage over its softness-matched network is only +0.011 (4/5 seeds), below our pre-registered threshold of +0.02, and no larger than that of a step-matched plain network (+0.010; BBB higher in 3/5 seeds) or a step-matched ensemble (+0.018; BBB higher in 1/5); standard-budget BBB does slightly better (+0.023). On the far-OOD sets of the same models long-trained BBB keeps an advantage (+0.047 and +0.034), though smaller than in the ten-class setting of Table 1 and Figure 1.

Table 4: Fashion-MNIST as in-distribution data, MNIST and KMNIST as OOD: Δ against the softness-matched 40-epoch DNN (mean over 5 seeds, seeds with $\Delta > 0$ in brackets; E21).

n	method	acc.	Δ MNIST	Δ KMNIST	MI Δ MNIST / KMNIST
2,000	DNN (reference)	0.827			
2,000	BBB, standard (4,800 steps)	0.831	+0.080 (5/5)	+0.066 (5/5)	+0.105 / +0.109
2,000	BBB, 23,550 steps	0.762	+0.136 (5/5)	+0.082 (5/5)	+0.015 / +0.205
2,000	DNN, 23,550 steps	0.823	+0.040 (5/5)	+0.018 (5/5)	
2,000	ensemble, 40 epochs	0.829	+0.038 (5/5)	+0.043 (5/5)	+0.099 / +0.112
2,000	ensemble, 23,550 steps	0.827	+0.085 (5/5)	+0.084 (5/5)	+0.092 / +0.094
10,000	DNN (reference)	0.863			
10,000	BBB, standard (23,550 steps)	0.849	+0.160 (5/5)	+0.206 (5/5)	+0.180 / +0.295

This is consistent with the mechanism in §6, which locates BBB's OOD signal in how inputs drive its first layer and which fails for near-OOD rotated digits, but we did not test the first-layer account on this split. The results in this paper therefore concern far-OOD detection.

4 An optimizer clock

Derivation. For one weight with posterior $\mathcal{N}(\mu, \sigma^2)$ and prior $\mathcal{N}(0, \sigma_p^2)$,

$$\mathrm{KL} = \log \frac{\sigma_p}{\sigma} + \frac{\sigma^2 + \mu^2}{2\sigma_p^2} - \frac{1}{2}, \qquad \frac{\partial\, \mathrm{KL}}{\partial \rho} = \left(\frac{\sigma}{\sigma_p^2} - \frac{1}{\sigma}\right) \mathrm{sigmoid}(\rho) \;\approx\; -1 \quad \text{for } \rho \ll 0, \tag{3}$$

because $\sigma = \mathrm{softplus}(\rho) \approx \mathrm{sigmoid}(\rho) \approx e^{\rho}$ when $\rho \ll 0$. In Eq. 1 this gradient is weighted by $1/N$, and while σ is small the likelihood term's gradient on ρ, which scales with σ, is negligible; the gradient on each ρ is then roughly the constant $-1/N$. Adam divides each gradient by a running estimate of its magnitude, so a gradient of constant sign produces a step of about the learning rate η, independently of its size and hence of N (Adam's sign-descent behaviour, Balles & Hennig, 2018). The number of steps for ρ to travel from ρ_0 to $\rho^\star$ is therefore

$$t^\star \approx \frac{\rho^\star - \rho_0}{\eta}. \tag{4}$$

For $\sigma = 0.3$ ($\rho^\star = -1.05$) from $\rho_0 = -5$ this gives $t^\star \approx 3{,}950$ steps at $\eta = 10^{-3}$. Under SGD with momentum β the step is instead $\eta_\rho/(N(1-\beta))$, so the same growth takes N times longer as N grows. The approximation fails once σ is no longer small compared with the prior, or once the likelihood pulls back. Unlu & Aitchison (2020) observe that mean-field variance parameters are often unconverged when training stops, and Nishida & Taiji (2022) report that BBB's variational parameters grow under Adam; Eq. 4 makes that growth quantitative and links it to OOD detection.

Test under Adam. We trained BBB with a constant learning rate for 40,000 steps at three learning rates and $n \in \{500, 10{,}000\}$ (E10, 3 seeds per arm; Figure 2, left). The time for the mean first-layer σ to reach 0.3 was 10,000, 5,000 and 3,000 steps at $n = 500$ and 12,000, 6,000 and 3,000 at $n = 10{,}000$ for $\eta = 5 \times 10^{-4}$, 10^{-3} and 2×10^{-3} (checkpoints every 1,000 steps, so the fastest arm is resolved only to about $\pm$1,000 steps): time $\times$ learning rate was constant to within the checkpoint resolution, and a twenty-fold change in n changed it by at most 20%. Eq. 4 predicts 7,900, 3,950 and 1,975 steps with no fitted constant. It gets the $1/\eta$ scaling and the independence from n right; the observed times are 1.27–1.52 times longer, consistent with gradient noise shrinking Adam's steps, and the gap is larger at $n = 10{,}000$, where the constant $1/N$ signal is weaker relative to the mini-batch noise. The onset of the OOD advantage roughly followed the same clock: onset $\times$ learning rate (in units of 10^3 steps $\times 10^{-3}$) was 5.5, 5.3 and 6.0 at $n = 10{,}000$, but 6.5, 7.7 and 10 at $n = 500$, where the largest learning rate later collapsed training (final accuracy 0.32).

Test under SGD. With SGD (momentum 0.9) on the ρ parameters, the mean σ after 20,000 steps was 0.67 at $n = 500$ and 0.017 at $n = 10{,}000$, a 39-fold difference (Figure 2, right). In the latter setting σ

never grew, and BBB had *no* OOD advantage (peak Δ -0.007) while reaching the best accuracy of all runs (0.966). With gradient clipping restricted to the means (E13), the SGD form of Eq. 4 predicts the numbers with no fitted constant: with $\eta_\rho = 0.05$ the step on ρ is about 5×10^{-5} at $n = 10{,}000$, giving $\sigma \approx 0.018$ after 20,000 steps against 0.0177 observed, and 10^{-3} at $n = 500$, giving $\sigma = 0.3$ after about 4,300 steps, against a first checkpoint above 0.3 at 5,000. The same optimizer then gives opposite OOD outcomes depending on whether σ grows: at $n = 500$, where it does, BBB reaches an advantage of $+0.057$ to $+0.060$ without clipping; at $n = 10{,}000$, where it does not, there is none. This within-optimizer contrast is the most direct causal evidence we have that the variance growth, not the optimizer or the data size as such, switches the advantage on. Unclipped SGD on ρ was unstable once σ grew large (6 of 9 runs diverged), which limits what the SGD arm can show beyond the growth phase (Appendix H).

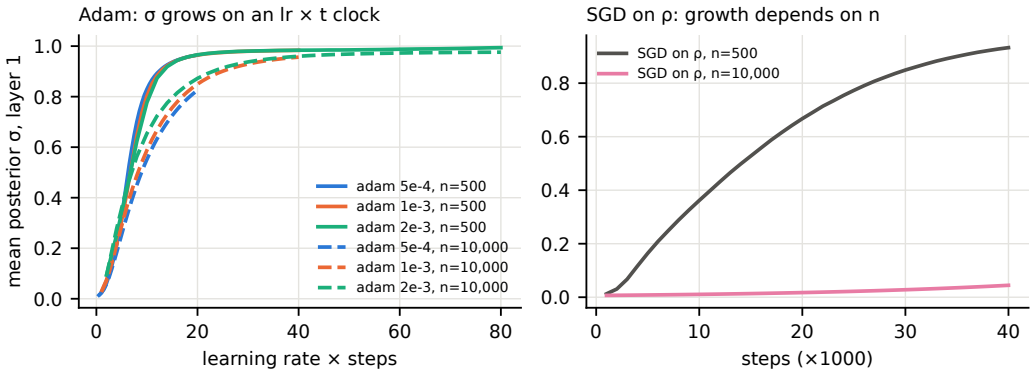


Figure 2: Mean first-layer posterior σ (mean of 3 seeds; E10). Left: Adam at three learning rates and two training-set sizes, against learning rate $\times$ steps. Right: SGD on the ρ parameters, against steps.

Natural-gradient VI escapes the clock, and showed no advantage. IVON keeps a Gaussian posterior $\mathcal{N}(m, 1/(N(h+\delta)))$ whose variance is set by N and a running curvature estimate h, not by a step count. In a pre-registered comparison (E14; 3 seeds per n, 40,000 steps), its mean first-layer σ was 0.170, 0.086 and 0.039 at $n = 500$, 2,000 and 10,000; the ratio of 4.41 between the extremes is close to the $1/\sqrt{N}$ value of 4.47, and the variances stayed far below the prior instead of drifting to it as BBB's did (0.96–0.98). At $n \geq 2{,}000$ IVON was as accurate as the plain network, but it had no advantage over the softness-matched network (peak Δ -0.005 and -0.008, against BBB's $+0.07$ and $+0.065$), in the one configuration we ran (learning rate and initial Hessian chosen once by validation NLL). Inflating its variances 3.1–9.3$\times$ by tempering the likelihood did not create one (E15, Appendix H).

Along training: entropy and mutual information diverge. Under the $\mathcal{N}(0, 1)$ prior the clock keeps running past the useful range (E6, Figure 3). At $n = 2{,}000$ with a constant learning rate, the entropy-based Δ on KMNIST peaks at $+0.067$ to $+0.074$ around 9,000–14,000 steps and is negative again by 25,000–30,000, as the model also becomes uncertain on in-distribution digits (mean test entropy rises from 0.30 to 0.94 as accuracy falls from 0.93 to 0.89), while the negative ELBO keeps improving. The MI-based Δ does not fall ($+0.071$ at 30,000 steps, $+0.061$ at 50,000). This drift is the posterior moving to an oversized prior: with $\sigma_p = 0.3$ (E11), σ plateaus at the prior scale by about 10,000 steps and both advantages persist (entropy Δ $+0.045$ and $+0.048$ at 40,000 steps, against -0.009 and $+0.015$ for $\sigma_p = 1$). Two cautions follow. A positive MI-based Δ does not by itself indicate a useful model: runs initialised with $\sigma_0 \geq 0.31$ never learn (chance accuracy) yet have the largest MI-based Δ (about $+0.10$). And neither held-out NLL nor the ELBO picks a good stopping point; a fixed step count does as well as a synthetic-outlier selection rule (E9, Appendix E), as the clock suggests.

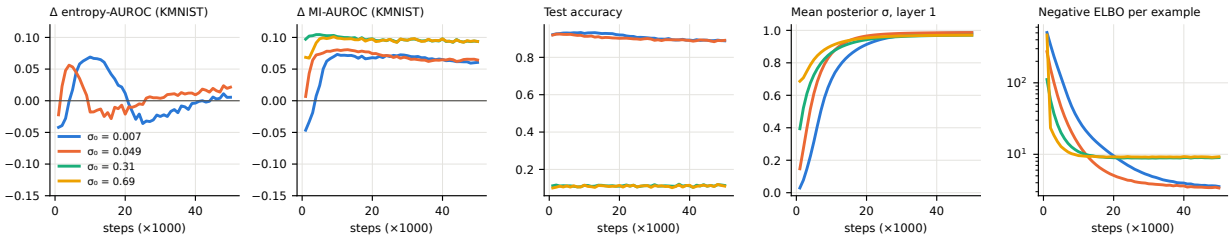


Figure 3: BBB training dynamics at $n = 2{,}000$ (constant learning rate; mean of 3 seeds per initial σ_0; 20 posterior samples; E6). Runs with $\sigma_0 \geq 0.31$ never learn; their entropy-based Δ (about -0.86) is below the plotted range.

5 The exact posterior

Is the advantage a property of the Bayesian posterior, or of mean-field VI under Adam? We compared BBB with full-batch Hamiltonian Monte Carlo (Neal, 2011) on the same model at width 100 (89,610 parameters), $n \in \{500, 2{,}000\}$. "Exact posterior" below means the posterior that HMC samples, as estimated from 800 samples per run with convergence checked on predictions, not on weights (which weight-space symmetries make uninformative). The softness-matched network here is the plain width-100 DNN, trained without a prior.

Under $\mathcal{N}(0, 1)$, HMC does not mix at this scale. With 4 chains of 2,500 iterations (E19), split-$\hat{R}$ on the predicted probability of the true class had a 90th percentile of 2.8 and 2.5 (threshold 1.1), so we draw no conclusion from that run. The obstacle is the posterior: its weights sit near the prior scale (standard deviation ≈ 0.85 in every layer), the logits reach $\sim$700, and the likelihood acts like a hard wall around "every training point correct". The stable leapfrog step fell from $\sim 3 \times 10^{-5}$ at 300-step trajectories to $\sim 1.5 \times 10^{-6}$ at 10^4 steps; a diagonal mass matrix and a narrower network did not help (pilots that looked only at sampler diagnostics; Appendix G).

Under $\mathcal{N}(0, 0.3^2)$ it converges, and the exact posterior has the advantage. A prior of scale 0.3 is where BBB's advantage is stable (§4). There the median stable step size was about 20–30 times larger at similar trajectory lengths. Eight chains per run (E20, E20b; identity mass matrix; dual-averaging step size (Hoffman & Gelman, 2014); trajectory length $\pi\sigma_p/2$ (Izmailov et al., 2021b); 500 warm-up and 1,000 sampling iterations, 100 kept samples per chain) passed the convergence check in all six runs (two values of n, three data subsets; 90th-percentile $\hat{R}$ 1.004–1.005, maximum 1.022). BBB was retrained under the same prior for 23,550 steps. The exact posterior beats the softness-matched network on both OOD sets on every run (Δ_K +0.052 to +0.077, Δ_F +0.074 to +0.115; Table 5, Figure 4; on the first subset FPR95 on KMNIST is 0.14–0.19 against 0.48–0.49) and is more accurate than BBB on every subset (mean 0.887 vs 0.784 at $n = 500$, 0.931 vs 0.906 at $n = 2{,}000$). BBB's MI-based advantage on KMNIST matches the exact posterior's on every subset (difference 0.002–0.012). Given the caution in §4 that MI rewards disagreement as such, this match is most meaningful at $n = 2{,}000$, where BBB is accurate (0.906) and also has a positive entropy-based advantage; at $n = 500$ BBB is much less accurate than HMC (0.784 against 0.887) and its entropy-based Δ is negative, so a matching MI there says less. Its entropy-based advantage recovers only part of it (mean Δ_K +0.030 against +0.075 at $n = 2{,}000$) and none at $n = 500$ (-0.032 against +0.060), and on Fashion-MNIST at $n = 2{,}000$ its MI falls short as well (mean +0.047 against +0.092).

Controls for the prior and the sample count. The softness-matched network has no prior, while HMC and BBB use $\mathcal{N}(0, 0.3^2)$. A pre-registered control (E23; Table 5) trained MAP networks and 5-member MAP ensembles under the same prior on the same subsets: the ensemble's advantage (Δ_K +0.024 and Δ_F +0.033 at $n = 2{,}000$, negative at $n = 500$) is below HMC's on every subset at both n, so HMC's advantage comes from posterior averaging, not from the prior plus ensembling. Evaluating BBB with 800 posterior samples, as many as HMC, changes its numbers by at most 0.01 at $n = 2{,}000$ but raises its entropy-based Δ by about 0.03 at $n = 500$ (to +0.000 and +0.012); 50 samples, used for BBB throughout, understate its entropy-based

Table 5: HMC posterior (8 chains × 100 samples per run) against BBB under the same $\mathcal{N}(0, 0.3^2)$ prior (width 100): mean and [range] over three data subsets (E20, E20b). On the first subset, FPR95 on KMNIST is 0.19 and 0.14 for HMC against 0.48–0.49 for the softened network.

n	method	acc.	Δ_F	Δ_K	MI Δ_F	MI Δ_K
500	HMC	0.887	+0.077 [+0.074, +0.079]	+0.060 [+0.051, +0.070]	+0.084	+0.065
500	BBB	0.784	−0.009 [−0.026, +0.001]	−0.032 [−0.054, −0.020]	+0.065	+0.065
2,000	HMC	0.931	+0.088 [+0.074, +0.115]	+0.075 [+0.074, +0.077]	+0.092	+0.078
2,000	BBB	0.906	+0.035 [+0.010, +0.077]	+0.030 [+0.027, +0.033]	+0.047	+0.071
Controls (E23), means over the same three subsets						
500	BBB, 800 samples	0.792	+0.012	+0.000	+0.076	+0.069
500	MAP ensemble, same prior	0.867	−0.053	−0.014	−0.020	−0.023
500	plain DNN (reference)	0.870				
2,000	BBB, 800 samples	0.908	+0.041	+0.039	+0.057	+0.077
2,000	MAP ensemble, same prior	0.929	+0.033	+0.024	+0.055	+0.035
2,000	plain DNN (reference)	0.917				

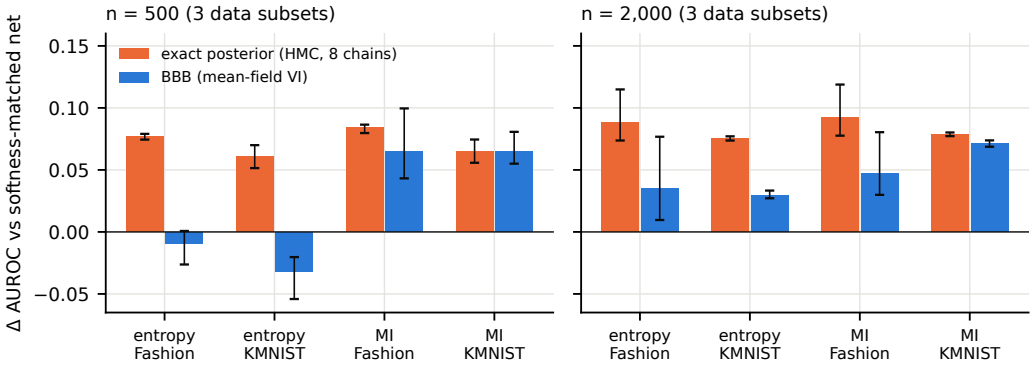


Figure 4: OOD advantage over the softness-matched network for the exact posterior (HMC) and BBB under the same $\mathcal{N}(0, 0.3^2)$ prior: mean over three data subsets, bars show the range (E20, E20b).

advantage at small n, while its MI-based Δ barely changes. With 800 samples BBB's MI-based Δ_K at $n = 2{,}000$ is +0.077, against HMC's +0.078.

Checks. On the first subset, a sampler self-test held: $\mathbb{E}[\theta \cdot \nabla U]/d$, which must equal 1 at stationarity, was 0.998–1.002 in every chain. Every single chain shows the advantage (Δ_K +0.064 to +0.073), so it is not diversity between chains, and a single posterior draw is poor (accuracy 0.72 and 0.83), so it comes from averaging. Part of it comes from HMC's higher accuracy: restricted to test images that both HMC and the plain network classify correctly, Δ is +0.054 to +0.099. An independent code review of the sampler and analysis found no error affecting these conclusions (Appendix G).

6 BBB gets there by a different route

The exact posterior's information is in correlations. We had predicted that the exact posterior would be less "overpruned" than BBB; it is the opposite (E20: 0% of hidden weights with |mean| > std within a chain, against 0.4–1.0% for BBB). Follow-up analyses (exploratory) show why. Each weight's marginal under the exact posterior is the prior: the within-chain standard deviation is 0.299–0.300 in every layer (prior 0.3) and |mean| is about 0.02, the sampling error of a mean of 100 draws. Drawing every weight independently from its HMC marginal gives chance accuracy (0.09). All of the exact posterior's information is in correlations between weights, which a mean-field family cannot hold. This is expected for a model with 89,610 parameters and at most 2,000 training points, since the data constrain only a small subspace; we report it as a confirmation that per-weight summaries cannot distinguish BBB from the exact posterior,

not as a discovery. (The median per-weight lag-1 autocorrelation between kept samples was −0.01, so 100 samples per chain are close to independent.)

BBB's information is in a few precise means. Long-trained BBB under $\mathcal{N}(0,1)$ is overpruned in the sense of Trippe & Turner (2017) (E16, Figure 6 in Appendix D): the median hidden-layer σ is 0.71–0.84 and only 1.2–4.5% of hidden weights have $|\mu| > \sigma$. A pre-registered swap test shows that both halves of this pairing matter: BBB's variances on the means of a step-matched plain network give 13–15% accuracy, and shuffling BBB's variances within each layer turns the OOD advantage negative (−0.16 to −0.32).

A first-layer noise-to-signal ratio, and its limits. For an input x, the first layer's mean pre-activation is $m_j = \sum_i \mu_{ij} x_i + \mu_{b_j}$ and the variance posterior noise adds to it is $v_j = \sum_i \sigma_{ij}^2 x_i^2 + \sigma_{b_j}^2$. Their ratio

$$R_1(x) = \frac{\sum_j v_j}{\sum_j m_j^2} \tag{5}$$

measures how much noise an input routes through the first layer relative to the signal it produces. Found in an exploratory analysis (E17) and confirmed in a separate pre-registered test on ten freshly trained BBB networks and two OOD sets never used before, grayscale CIFAR-10 and SVHN (E18, Figure 5), R_1 reached AUROC 0.960–0.991 on all four far-OOD sets, within 0.003–0.019 of BBB's 50-sample mutual information, and correlated with BBB's per-image MI (Spearman 0.71–0.83). But on these far-OOD sets, where everything is near the ceiling, a pre-registered comparison with input-only statistics (E23) shows that this does not single out a mechanism: the pixel rule of §3 correlates with BBB's per-image MI at least as well as R_1 does ($|\rho|$ 0.77–0.87 against 0.71–0.83), so we cannot claim that R_1 explains BBB's far-OOD signal better than where the ink is. (Resetting those pixels does not change BBB's advantage, so BBB does not depend on them; they are correlated with what it detects.) R_1 does carry information beyond input statistics on near-OOD rotated digits, where it reaches AUROC 0.86–0.88 while the image's total ink, its ink count and the pixel rule are at chance (0.50–0.55); there BBB's own MI is better still (0.950 against 0.863 at $n = 10{,}000$), so deeper layers add information. Computed from the exact posterior's per-weight moments instead, R_1 separates OOD inputs only weakly (0.69–0.88): it describes BBB's mean-field solution, not the exact posterior. R_1 is a first-layer instance of the moment propagation used in the local reparameterisation trick (Kingma et al., 2015) and in sampling-free variational inference (Wu et al., 2019; Haußmann et al., 2019; Postels et al., 2019), and a cheap sampling-free score for a trained BBB network; we do not claim it as a full account of BBB's OOD behaviour.

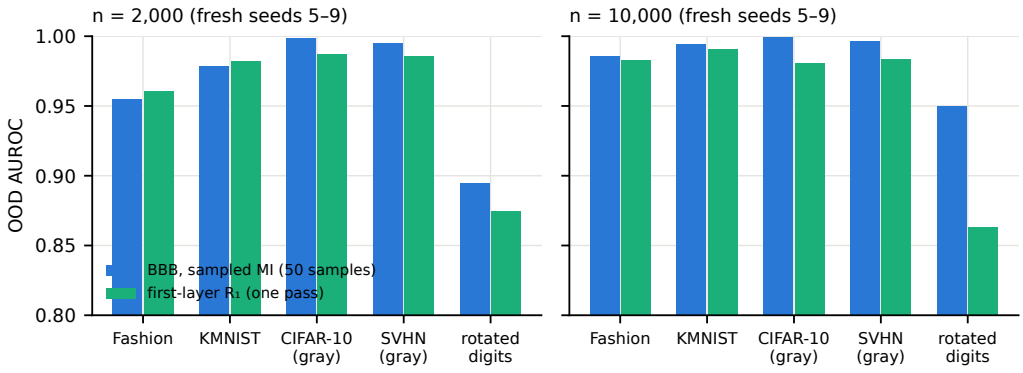


Figure 5: OOD AUROC of BBB's sampled mutual information and of the analytic first-layer ratio R_1 (Eq. 5) on freshly trained models (mean of 5 seeds; E18). CIFAR-10 and SVHN were not used anywhere else in the study.

IVON in this light. IVON's per-weight standard deviations (0.04–0.17 under $\mathcal{N}(0,1)$) are far below the spread of exact-posterior samples, which sits at the prior scale under $\mathcal{N}(0, 0.3^2)$ and had already reached

≈ 0.85 in the unconverged $\mathcal{N}(0,1)$ chains. Its data-scaled variances are therefore not closer to the exact marginals than BBB's; what BBB shares with the exact posterior is not its marginals but the size of its epistemic signal.

7 Related work

Softness, temperature and OOD baselines. Temperature scaling (Guo et al., 2017) and its role in OOD detection (Liang et al., 2018) motivate the control; Ovadia et al. (2019) include temperature-scaled baselines and find ensembles strongest under shift, Ashukha et al. (2020) argue for comparing uncertainty methods after temperature scaling, and Abe et al. (2022) find that ensembles' OOD gains are largely explained by their in-distribution performance. Our control extends these to a per-method entropy match for OOD detection. The deterministic detectors in Table 2 are the maximum softmax probability (Hendrycks & Gimpel, 2017), the energy score (Liu et al., 2020), a Mahalanobis detector (Lee et al., 2018) and deep nearest neighbours (Sun et al., 2022); feature-density methods such as DDU (Mukhoti et al., 2023) are a stronger deterministic family we did not test.

Mean-field VI and its variances. Trippe & Turner (2017) describe overpruning; Farquhar et al. (2020) argue that deep mean-field networks can be expressive enough in practice; Coker et al. (2022) prove that the optimal mean-field predictive approaches the prior predictive as width grows, consistent with the drift in §4; Foong et al. (2020) show what mean-field approximations cannot represent; Odgers et al. (2026) show that mean-field VI can overestimate in-distribution predictive variance. Unlu & Aitchison (2020) and Nishida & Taiji (2022) observe unconverged or growing variance parameters under standard optimizers; we derive the N-independent clock behind this growth and link it to OOD detection. Natural-gradient methods such as Vadam (Khan et al., 2018), VOGN (Osawa et al., 2019) and IVON (Shen et al., 2024) tie the variances to N and the curvature by construction.

Exact posteriors and priors. Izmailov et al. (2021b) run full-batch HMC on much larger networks and find that the exact posterior predicts well and differs from mean-field VI; we find the same split for OOD detection at small scale, and that an $\mathcal{N}(0,1)$ prior makes HMC far harder than a prior of scale 0.3. Izmailov et al. (2021a) show that weights on always-zero pixels keep their prior under HMC and hurt accuracy under covariate shift, and Kristiadi et al. (2020) show that a Bayesian last layer reduces overconfidence far from the data, which our last-layer Laplace results do not reproduce for these OOD sets. Henning et al. (2021) argue that whether exact posteriors are good OOD detectors depends on the prior and architecture; we give one setting where the exact posterior beats a softness-matched network. Fortuin et al. (2022) show that isotropic Gaussian priors are often poor choices, Wenzel et al. (2020) that tempered posteriors can predict better than the Bayes posterior, and Depeweg et al. (2018) and Wimmer et al. (2023) discuss the entropy/mutual-information decomposition our two scores rest on. Raina & Schmah (2025) report Bayesian advantages for OOD detection from small training sets; our results suggest checking training length and softness before attributing such gains to the posterior.

8 Limitations

Scale and data. All positive results come from small MLPs on MNIST-family data (MNIST or Fashion-MNIST as in-distribution), on far-OOD pairs that a model-free pixel rule nearly solves; on a held-out-class split the advantage largely disappears. On CIFAR-10 with small, underfit models (a CNN and the same MLP, 37–68% accuracy, no augmentation) no comparable advantage appeared (Appendix F); this says little about well-trained networks, and whether the clock and the advantage carry over to modern architectures is open.

Exact posterior. HMC converged only under the $\mathcal{N}(0, 0.3^2)$ prior, at width 100, on three data subsets per n; under BBB's default $\mathcal{N}(0,1)$ prior it did not mix, so the comparison with the exact posterior does not cover the setting of most other experiments. There is one HMC run and one BBB model per subset, the $\hat{R}$

used is not rank-normalised, and at $n = 2{,}000$ the potential drifted slightly between the halves of sampling (2.2 standard errors).

Design. Only three prior scales were tried, at one training-set size; the dynamics runs use a constant learning rate and 3 seeds, the fixed-budget runs a cosine schedule; the clock was tested with Adam and SGD on one architecture, with timings resolved to 1,000-step checkpoints; all methods use default hyperparameters without per-method tuning, except IVON's learning rate and initial Hessian (chosen once by validation NLL); most comparisons use 5 seeds (3 for the dynamics); and the pre-registration records are self-attested.

9 Conclusion

For Bayes by Backprop on small MLPs and far-OOD MNIST-family benchmarks, "knowing what it doesn't know" is not a fixed property of the method: on MNIST it is set by optimizer time. Under Adam the posterior variances grow on a learning-rate × steps clock, almost regardless of the amount of data, and the OOD advantage over a softness-matched network switches on with them; where the variances stay small, as under SGD with more data, there is none. The advantage is not only an artefact of mean-field VI: under a prior where it can be sampled, the HMC posterior has an advantage over the softness-matched network, and on KMNIST it is the same size as BBB's mutual-information signal. But BBB reaches that signal by a structurally different solution, a few precise means with prior-scale noise elsewhere, rather than by approximating the exact posterior's marginals. Comparisons of Bayesian and deterministic OOD detectors should report the optimizer, learning rate and training budget, the uncertainty score, and a softness-matched baseline.

Broader Impact Statement

This is a methodological study of uncertainty estimates on public image benchmarks (MNIST, Fashion-MNIST, KMNIST, CIFAR-10, CIFAR-100, SVHN); it involves no human subjects and no personal data. Its main societal relevance is a caution: OOD detection is used to decide when a model's prediction should not be trusted, and our results show that a Bayesian network's apparent OOD advantage can depend on the training run, can be matched or exceeded by simple baselines, and can vanish on near-OOD inputs. Over-reading such advantages could lead to unwarranted trust in deployed systems; we mitigate this by reporting softness-matched and step-matched baselines, a near-OOD split and all falsified predictions, and by recommending that these be reported routinely. All experiments ran on two consumer GPUs; the method itself poses no apparent risk of misuse.

Reproducibility. Code, every command, and every pre-registered protocol with its result are provided as anonymised supplementary material.

References

Taiga Abe, E. Kelly Buchanan, Geoff Pleiss, Richard Zemel, and John P. Cunningham. Deep ensembles work, but are they necessary? In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

Arsenii Ashukha, Alexander Lyzhov, Dmitry Molchanov, and Dmitry Vetrov. Pitfalls of in-domain uncertainty estimation and ensembling in deep learning. In *International Conference on Learning Representations (ICLR)*, 2020.

Lukas Balles and Philipp Hennig. Dissecting Adam: The sign, magnitude and variance of stochastic gradients. In *International Conference on Machine Learning (ICML)*, 2018.

Charles Blundell, Julien Cornebise, Koray Kavukcuoglu, and Daan Wierstra. Weight uncertainty in neural networks. In *International Conference on Machine Learning (ICML)*, 2015.

Tarin Clanuwat, Mikel Bober-Irizar, Asanobu Kitamoto, Alex Lamb, Kazuaki Yamamoto, and David Ha. Deep learning for classical Japanese literature. *arXiv preprint arXiv:1812.01718*, 2018.

Beau Coker, Wessel P. Bruinsma, David R. Burt, Weiwei Pan, and Finale Doshi-Velez. Wide mean-field Bayesian neural networks ignore the data. In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022.

Erik Daxberger, Agustinus Kristiadi, Alexander Immer, Runa Eschenhagen, Matthias Bauer, and Philipp Hennig. Laplace redux – effortless Bayesian deep learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

Stefan Depeweg, José Miguel Hernández-Lobato, Finale Doshi-Velez, and Steffen Udluft. Decomposition of uncertainty in Bayesian deep learning for efficient and risk-sensitive learning. In *International Conference on Machine Learning (ICML)*, 2018.

Sebastian Farquhar, Lewis Smith, and Yarin Gal. Liberty or depth: Deep Bayesian neural nets do not need complex weight posterior approximations. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

Andrew Y. K. Foong, David R. Burt, Yingzhen Li, and Richard E. Turner. On the expressiveness of approximate inference in Bayesian neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

Vincent Fortuin, Adrià Garriga-Alonso, Sebastian W. Ober, Florian Wenzel, Gunnar Rätsch, Richard E. Turner, Mark van der Wilk, and Laurence Aitchison. Bayesian neural network priors revisited. In *International Conference on Learning Representations (ICLR)*, 2022.

Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *International Conference on Machine Learning (ICML)*, 2016.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, 2017.

Manuel Haußmann, Fred A. Hamprecht, and Melih Kandemir. Sampling-free variational inference of Bayesian neural networks by variance backpropagation. In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2019.

Dan Hendrycks and Kevin Gimpel. A baseline for detecting misclassified and out-of-distribution examples in neural networks. In *International Conference on Learning Representations (ICLR)*, 2017.

Christian Henning, Francesco D'Angelo, and Benjamin F. Grewe. Are Bayesian neural networks intrinsically good at out-of-distribution detection? In *ICML Workshop on Uncertainty and Robustness in Deep Learning (UDL)*, 2021.

Matthew D. Hoffman and Andrew Gelman. The no-U-turn sampler: Adaptively setting path lengths in Hamiltonian Monte Carlo. *Journal of Machine Learning Research*, 15:1593–1623, 2014.

Pavel Izmailov, Patrick Nicholson, Sanae Lotfi, and Andrew Gordon Wilson. Dangers of Bayesian model averaging under covariate shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021a.

Pavel Izmailov, Sharad Vikram, Matthew D. Hoffman, and Andrew Gordon Wilson. What are Bayesian neural network posteriors really like? In *International Conference on Machine Learning (ICML)*, 2021b.

Mohammad Emtiyaz Khan, Didrik Nielsen, Voot Tangkaratt, Wu Lin, Yarin Gal, and Akash Srivastava. Fast and scalable Bayesian deep learning by weight-perturbation in Adam. In *International Conference on Machine Learning (ICML)*, 2018.

Durk P. Kingma, Tim Salimans, and Max Welling. Variational dropout and the local reparameterization trick. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.

Agustinus Kristiadi, Matthias Hein, and Philipp Hennig. Being Bayesian, even just a bit, fixes overconfidence in ReLU networks. In *International Conference on Machine Learning (ICML)*, 2020.

Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

Kimin Lee, Kibok Lee, Honglak Lee, and Jinwoo Shin. A simple unified framework for detecting out-of-distribution samples and adversarial attacks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

Shiyu Liang, Yixuan Li, and R. Srikant. Enhancing the reliability of out-of-distribution image detection in neural networks. In *International Conference on Learning Representations (ICLR)*, 2018.

Weitang Liu, Xiaoyun Wang, John Owens, and Yixuan Li. Energy-based out-of-distribution detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.

Jishnu Mukhoti, Andreas Kirsch, Joost van Amersfoort, Philip H. S. Torr, and Yarin Gal. Deep deterministic uncertainty: A new simple baseline. In *Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

Radford M. Neal. MCMC using Hamiltonian dynamics. In *Handbook of Markov Chain Monte Carlo*. Chapman and Hall/CRC, 2011.

Keigo Nishida and Makoto Taiji. AdamB: Decoupled Bayes by backprop with Gaussian scale mixture prior. *IEEE Access*, 10:92959–92970, 2022.

James Odgers, Ben Riegler, Siddharth Swaroop, and Vincent Fortuin. Gaussian mean field variational inference can overestimate predictive variance. In *International Conference on Machine Learning (ICML)*, 2026.

Kazuki Osawa, Siddharth Swaroop, Anirudh Jain, Runa Eschenhagen, Richard E. Turner, Rio Yokota, and Mohammad Emtiyaz Khan. Practical deep learning with Bayesian principles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

Janis Postels, Francesco Ferroni, Huseyin Coskun, Nassir Navab, and Federico Tombari. Sampling-free epistemic uncertainty estimation using approximated variance propagation. In *International Conference on Computer Vision (ICCV)*, 2019.

Kevin Raina and Tanya Schmah. Out-of-distribution detection from small training sets using Bayesian neural network classifiers. In *British Machine Vision Conference (BMVC)*, 2025.

Yuesong Shen, Nico Daheim, Bai Cong, Peter Nickl, Gian Maria Marconi, Clement Bazan, Rio Yokota, Iryna Gurevych, Daniel Cremers, Mohammad Emtiyaz Khan, and Thomas Möllenhoff. Variational learning is effective for large deep networks. In *International Conference on Machine Learning (ICML)*, 2024.

Yiyou Sun, Yifei Ming, Xiaojin Zhu, and Yixuan Li. Out-of-distribution detection with deep nearest neighbors. In *International Conference on Machine Learning (ICML)*, 2022.

Brian Trippe and Richard Turner. Overpruning in variational Bayesian neural networks. In *NIPS Workshop on Advances in Approximate Bayesian Inference*, 2017.

Ali Unlu and Laurence Aitchison. Variational Laplace for Bayesian neural networks. *arXiv preprint arXiv:2011.10443*, 2020.

Florian Wenzel, Kevin Roth, Bastiaan S. Veeling, Jakub Świątkowski, Linh Tran, Stephan Mandt, Jasper Snoek, Tim Salimans, Rodolphe Jenatton, and Sebastian Nowozin. How good is the Bayes posterior in deep neural networks really? In *International Conference on Machine Learning (ICML)*, 2020.

Lisa Wimmer, Yusuf Sale, Paul Hofman, Bernd Bischl, and Eyke Hüllermeier. Quantifying aleatoric and epistemic uncertainty in machine learning: Are conditional entropy and mutual information appropriate measures? In *Conference on Uncertainty in Artificial Intelligence (UAI)*, 2023.

Anqi Wu, Sebastian Nowozin, Edward Meeds, Richard E. Turner, José Miguel Hernández-Lobato, and Alexander L. Gaunt. Deterministic variational inference for robust Bayesian neural networks. In *International Conference on Learning Representations (ICLR)*, 2019.

Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-MNIST: a novel image dataset for benchmarking machine learning algorithms. *arXiv preprint arXiv:1708.07747*, 2017.

A KL scaling

Dividing the KL term in Eq. 1 by the number of mini-batches instead of by N is an easy slip when the cross-entropy is averaged per batch. It over-weights the KL term by $N/\lceil N/B \rceil \approx B$ (62.5× at $N = 2{,}000$, $B = 64$). In a 3-seed pilot it cost BBB 3.2 points of accuracy (0.904 vs 0.936) and raised ECE from 0.022 to 0.185. On accuracy, correctly scaled BBB beats a MAP network trained with the same prior and budget by 1.1 points at $n = 500$ and 0.5 points at $n = 2{,}000$ (9/10 and 10/10 seeds), and loses by 0.5 points at $n = 10{,}000$ (E3).

B Always-background pixels

On average 192, 148 and 104 pixels are constant (always background) across a training subset at $n = 500$, 2,000 and 10,000. Flagging any image with ink on such a pixel separates MNIST from Fashion-MNIST and KMNIST with AUROC 0.967–1.000 over 15 training subsets (means 0.990–1.000). Resetting those pixels to background in the OOD images leaves every method's AUROC and Δ essentially unchanged (BBB at $n = 10{,}000$: $+0.057 \to +0.055$ and $+0.070 \to +0.070$), and the growth of σ is global rather than concentrated on those pixels (ratio 1.0–1.24).

C Which layers' noise matters

Taking BBB models trained for 23,550 steps, we switched posterior noise off layer by layer (E6b). Noise in the first and second layers together keeps 72–82% of Δ; noise in any single layer keeps essentially none (first layer: 16% on Fashion-MNIST, -26% on KMNIST), and the posterior-mean network on its own is a worse OOD detector than the DNN. Noise in the last layer alone does not reproduce the advantage in this model, which may explain why last-layer Laplace does poorly in Table 1.

D BBB's overpruned posterior

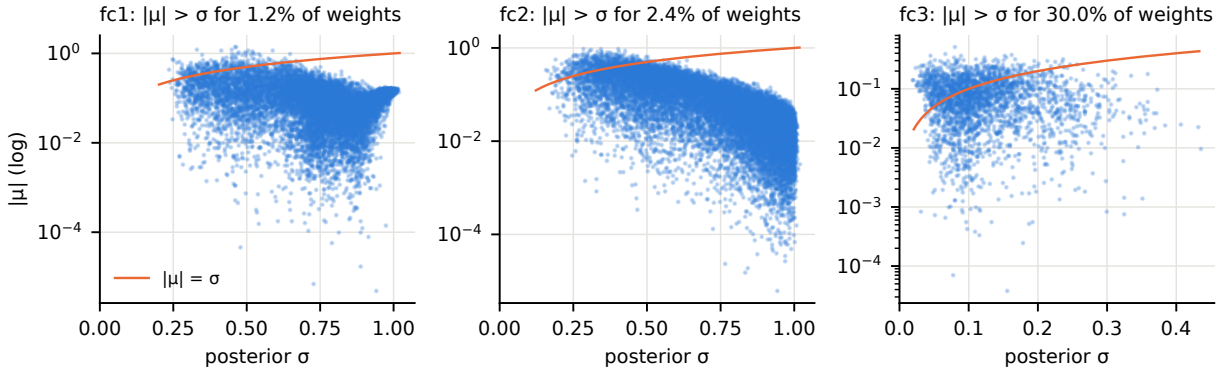


Figure 6: Posterior mean magnitude against posterior σ for every weight of a BBB network trained for 23,550 steps ($n = 2{,}000$, seed 0; 20,000 weights sampled per layer; E16). In the second layer $|\mu|$ and σ are strongly anti-correlated (-0.64 to -0.72), and the last layer is nearly deterministic.

E Choosing when to stop

Minimum held-out NLL stops at 1,000–3,000 steps, before the advantage appears; minimum −ELBO stops at the end of training. A rule that needs no real OOD data, picking the checkpoint that best separates validation images from three synthetic outlier sets (uniform noise, flipped digits, pixel-permuted digits) by MI, reaches 74% of the best achievable entropy-based advantage (an optimistic oracle over 40 checkpoints), but simply stopping at a fixed 8,000 steps reaches 72% (Table 6; 12 runs, $n \in \{500, 2{,}000, 10{,}000\}$; E9). Scoring the last checkpoint by MI instead of entropy, with no selection at all, does as well (+0.049).

Table 6: Stopping rules ($\bar{\Delta}$ = mean of Fashion-MNIST and KMNIST entropy-based Δ at the chosen checkpoint; 12 runs; E9).

rule	mean step	$\bar{\Delta}$ (entropy)	fraction of oracle
oracle (uses real OOD; optimistic)	10,917	+0.067	1.00
synthetic-outlier proxies (MI)	10,417	+0.049	0.74
fixed stop at 8,000 steps	8,000	+0.048	0.72
fixed stop at 10,000 steps	10,000	+0.047	0.71
minimum −ELBO	39,417	+0.034	0.51
minimum validation NLL	2,250	−0.032	−0.48

F CIFAR-10

We repeated the dynamics experiment on CIFAR-10 ($n = 10{,}000$, no augmentation) with a small CNN (three 3×3 convolutional layers of 32, 64 and 128 channels, then 256 and 10 units), every layer Bayesian, with SVHN and CIFAR-100 as OOD sets (E7). No comparable advantage appeared (Figure 7): BBB's peak Δ on SVHN was -0.044, $+0.006$ and -0.056 in the three seeds, and on CIFAR-100 at most $+0.031$, early in training. Even a 5-member deep ensemble lost to a single softness-matched DNN on SVHN (AUROC 0.727–0.740 vs 0.759–0.807). The drift towards the prior did replicate: the fully connected layer's σ reached 0.996 and accuracy fell from 0.67–0.68 to 0.61–0.63. The MNIST study's MLP on flattened CIFAR-10 images (E7b; 38% and 46% deterministic accuracy at $n = 2{,}000$ and 10,000) also showed no advantage on SVHN (peak Δ -0.031 to $+0.030$); its CIFAR-100 gains ($+0.030$ to $+0.046$ early in training) were similar to the CNN's, and larger "gains" appeared only after collapse to near-chance accuracy. These models are small and underfit, so this negative result says little about well-trained networks.

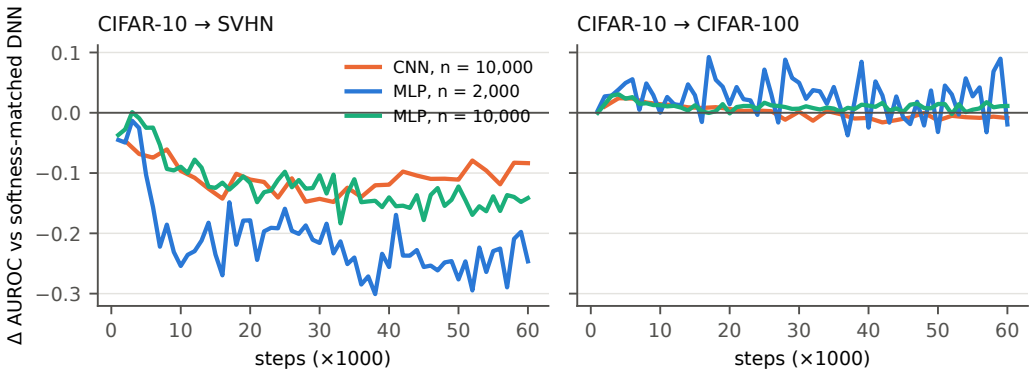


Figure 7: BBB against its softness-matched network on CIFAR-10 over training (mean of 3 seeds per curve; E7, E7b).

G HMC details

Sampler. Full-batch HMC in PyTorch, with all chains of a run batched in one process (the per-chain implementation was limited by Python overhead, not by the GPU). Each chain starts from its own MAP estimate (3,000 full-batch Adam steps from a different random initialisation), uses an identity mass matrix, adapts its step size by dual averaging towards 0.8 acceptance over 500 warm-up iterations and then freezes it (with ±10% jitter), and uses $L = \lceil 0.471/\text{median step} \rceil$ leapfrog steps (360–430 at $n = 500$, 702–856 at $n = 2{,}000$). Every tenth of 1,000 sampling iterations is kept. A run takes about 25 minutes ($n = 500$) and 55 minutes ($n = 2{,}000$) on one RTX 3090.

Convergence. Split-$\hat{R}$ over the 8 chains on the predicted probability of the true class for 1,000 fixed test images; a run counts as converged if the 90th percentile is below 1.1 (decided before running). The mean potential changed by -1.9 and $+11.6$ between the halves of sampling on the first subset (batch-means standard errors 5.5 and 5.3). The self-test $\mathbb{E}[\theta \cdot \nabla U] = d$ holds for any differentiable target at stationarity; a wrong gradient, kinetic energy or acceptance step would violate it.

The $\mathcal{N}(0,1)$ attempt (E19). 4 chains, 300 leapfrog steps, Robbins–Monro step-size adaptation over 1,500 warm-up iterations, 1,000 sampling iterations. The adapted step size (2.3–5.4×10^{-5}) moved each chain only $\sim$0.01 per iteration, the potential was still rising during sampling, and the spread of weight means across chains was 6–8× the within-chain standard deviation. Under $\mathcal{N}(0,1)$ the stable step shrinks as trajectories lengthen, so longer runs would not fix this at an affordable cost; a diagonal mass matrix estimated in Stan-style windows reduced the step size further, and a width-20 network gave a step of 10^{-5} at 3,000 leapfrog steps.

Independent code review. A separate AI agent (a large language model run independently of the analysis, with read-only access to the code and results) audited the sampler and analysis code (kinetic energy and acceptance, state after accept/reject, state-independent step size and trajectory length, exclusion of warm-up samples, $\hat{R}$ axes, validation-only temperature fit, identical data subsets for HMC, BBB and the plain network) and recomputed Δ independently; it found no error affecting the conclusions. Its suggestions (accuracy-matched Δ, drift standard errors, seed spread) are reported above.

H Pre-registered predictions and outcomes

Each protocol was written before its runs: committed to version control for E1–E9 and, for E10–E23, with a SHA-256 hash recorded before launch (self-attested, not independently timestamped). Of the 87 entries below, 39 were confirmed, 31 falsified, 13 had mixed outcomes, 3 had no outcome and 1 was a decision rule.

(E5's first prediction held by construction and is omitted.) The falsified ones are what moved the study from "is the advantage just softness?" to the training-path account.

exp.	prediction	outcome
E1	P0: new pipeline reproduces old numbers (± 0.02 acc, ± 0.03 AUROC)	acc/ECE yes; AUROC off by 0.035
E1	P1: correct KL weight raises BBB accuracy	confirmed
E1	P2: BBB within 0.02 AUROC of softness-matched DNN	falsified (Fashion)
E2	P1: BBB never beats the control	falsified ($n = 10{,}000$)
E2	P2: ensembles beat the control	confirmed
E2	P3: MC dropout and Laplace within 0.02 of the control	dropout yes; Laplace falsified
E2	P4: larger KL weight lowers accuracy, raises KMNIST MI-AUROC	confirmed
E3	P1: BBB and budget-matched MAP equally accurate (< 0.005)	falsified (narrowly)
E3	P2: BBB never beats softness-matched MAP	falsified ($n = 10{,}000$)
E4	P1: the gain follows optimizer steps, not n	confirmed
E4	P2: dead-pixel weights have $> 5\times$ the σ of live ones	falsified
E4	P3: $n = 500$ with long training shows the gain	confirmed
E5	P2: removing dead-pixel ink halves BBB's gain	falsified
E5	P3: removing it shrinks the ensemble's gain	falsified
E5	P4: the DNN's AUROC drops less than BBB's when the ink is removed	falsified (neither drops)
E6	P1: Δ rises with σ	falsified (peaks, then falls)
E6	P2: Δ plateaus	falsified (transient)
E6	P3: larger σ_0 reaches the gain sooner	falsified ($\sigma_0 \geq 0.31$ never learns)
E6	P4: ELBO still improving at 4,800 steps	confirmed
E6b	P1: first-layer noise alone keeps $\geq 70\%$	falsified
E6b	P2: last-layer noise alone keeps $< 30\%$	confirmed
E9	P1: combined proxies reach $\geq 70\%$ of oracle; NLL $\leq 20\%$	confirmed, but a fixed 8k-step stop does as well
E9	P2: one proxy alone reaches $\geq 70\%$	falsified (67%)
E9	P3: peak step rises with n	confirmed by mean; not robust
E7	P1: window on SVHN with a CNN	falsified
E7	P2: window closes	no outcome (no window)
E7	P3: proxy selection beats NLL	falsified
E7	P4: ensemble beats the control on SVHN	falsified
E7b	rule: architecture vs data	"architecture" by the rule; weak
E10	P1: time to $\sigma = 0.3 \times$ learning rate constant ($\pm 30\%$)	confirmed
E10	P2: under Adam, n changes that time by $\leq 25\%$	confirmed
E10	P3: under SGD, σ grows $\geq 3\times$ faster at $n = 500$ than 10,000	confirmed ($39\times$)
E10	P4: onset of the OOD advantage follows the same clock ($\pm 40\%$)	confirmed
E11	P1: a tighter prior improves final accuracy	falsified
E11	P2: a tighter prior keeps the entropy advantage to the end	$\sigma_p = 0.3$ yes; 0.1 no
E11	P3: MI advantage survives $\sigma_p \in \{0.1, 0.3\}$ (kill test)	confirmed
E12	P1: weight noise stays ≥ 0.02 below BBB	confirmed
E12	P2: step-matched softened network stays ≥ 0.04 below BBB	confirmed (narrowly)
E12	P3: step-matched ensemble stays below BBB	confirmed
E12	P4: feature detectors ≥ 0.95 and within 0.03 of BBB	kNN yes; Mahalanobis no
E13	P1: without joint clipping, SGD σ still grows $\geq 3\times$ faster at $n = 500$	confirmed ($53\times$, one $n = 500$ seed past 20k)
E13	P2: ρ learning rate $\propto N$ aligns the clocks across n	falsified as scored (1 run diverged first; 2 crossed just before diverging)
E13	P3: with that scaling, the OOD advantage appears at $n = 10{,}000$	falsified (runs diverged first)
E14	P1: IVON's σ is $\geq 2\times$ larger at $n = 500$ than 10,000	confirmed ($4.41\times$)

exp.	prediction	outcome
E14	P2: IVON's $\sigma \leq 0.3$ at $n \geq 2{,}000$	confirmed
E14	P3: IVON shows no transient	$n \geq 2{,}000$ yes; $n = 500$ no
E14	P4: IVON's advantage below BBB's but ≥ 0	below BBB yes; ≥ 0 no
E14	P5: IVON accuracy within 0.01 of the plain network	$n \geq 2{,}000$ yes; $n = 500$ no
E15	P1: tempering raises IVON's σ	confirmed
E15	P2: at $k = 100$, IVON gains an advantage ≥ 0.03	falsified (collapse)
E15	P3: at $k = 10$, accuracy within 0.02	$n = 10{,}000$ yes; $n = 2{,}000$ no
E16	P1: BBB's σ on plain-network means collapses accuracy	confirmed
E16	P2: that swap loses ≥ 0.05 of the advantage	confirmed
E16	P3: shuffling σ within layers keeps $\geq 80\%$	falsified
E17	P1: second-layer ratio R_2 separates OOD (AUROC ≥ 0.90)	falsified
E17	P2: R_2 correlates with MI (≥ 0.6)	falsified
E17	P3: R_2 separates better than R_1	falsified
E18	P1: R_1 AUROC ≥ 0.90 on four far-OOD sets (fresh models)	confirmed
E18	P2: R_1 within 0.03 of BBB's MI	confirmed
E18	P3: Spearman(R_1, MI) ≥ 0.6	confirmed
E18	P4: R_1 within 0.05 of MI on rotated digits	$n = 2{,}000$ yes; $n = 10{,}000$ no
E19	P1: decision rule on HMC's Δ (exact posterior)	no outcome: chains did not converge ($\hat{R}$ 90th pct. 2.5–2.8)
E19	P2: HMC less overpruned than BBB ($\geq 2\times$)	no outcome: within-chain spread reflects poor mixing
E19	P3: HMC accuracy $\geq$ BBB at $n \in \{500, 2000\}$	falsified (0.899 vs 0.920; 0.833 vs 0.881), exploratory
E20	P1: decision rule on HMC's Δ under $\mathcal{N}(0, 0.3^2)$ (converged)	decision: advantage at both n
E20	P2: HMC less overpruned than BBB ($\geq 2\times$)	falsified (0% vs 0.4–1.0%)
E20	P3: HMC accuracy $\geq$ BBB	confirmed (0.890 vs 0.798; 0.929 vs 0.909)
E20	P4: HMC and BBB MI Δ_K within 0.03	confirmed (0.006 at both n)
E20b	Q1: HMC $\Delta_K, \Delta_F \geq +0.03$ on every subset	confirmed (6/6 runs)
E20b	Q2: mean $\lvert$MI Δ_K HMC $-$ BBB$\rvert \leq 0.03$	confirmed (0.006, 0.007)
E20b	Q3: HMC accuracy $\geq$ BBB on every subset	confirmed (6/6)
E20b	Q4: HMC entropy Δ_K exceeds BBB's by ≥ 0.02	confirmed (+0.092, +0.045)
E21	P1: Fashion-MNIST ID, standard-budget BBB $\Delta \leq +0.02$	falsified (+0.080, +0.066)
E21	P2: long-trained BBB $\Delta \geq +0.03$, $\geq 4/5$ seeds, both OOD sets	confirmed (+0.136, +0.082; 5/5)
E21	P3: long BBB beats step-matched DNN and ensemble ($\geq 4/5$ seeds)	DNN yes (5/5, 5/5); ensemble MNIST 5/5, KMNIST 2/5
E21	P4: $n = 10{,}000$ BBB $\Delta \geq +0.03$	confirmed (+0.160, +0.206)
E21	P5: long BBB MI $\Delta \geq +0.03$ on both OOD sets	KMNIST yes (+0.205); MNIST no (+0.015)
E22	P1: digits 0–4 ID; long BBB $\Delta \geq +0.02$ on held-out digits 5–9	falsified (+0.011, 4/5)
E22	P2: held-out-digit Δ below far-OOD Δ	confirmed (+0.011 vs +0.041)
E22	P3: long BBB beats step-matched DNN and ensemble on held-out digits	falsified (3/5, 1/5)
E22	P4: pixel rule AUROC on held-out digits ≤ 0.80	confirmed (0.569)
E22	P5: standard-budget BBB held-out $\Delta \leq +0.02$	falsified (+0.023)
E23	P1: HMC Δ exceeds a MAP ensemble's under the same prior, every subset	confirmed (3/3 at both n)
E23	P2: single MAP under the same prior, mean $\Delta_K \leq +0.02$	confirmed (-0.015, +0.017)
E23	P3: BBB with 800 vs 50 samples changes Δ by ≤ 0.01	$n = 2{,}000$ yes; $n = 500$ no (+0.032)
E23	P4: R_1 tracks BBB's MI better than input-only baselines (far OOD)	falsified (pixel rule $\lvert\rho\rvert$ 0.77–0.87 vs 0.71–0.83)
E23	P5: R_1 beats input-only baselines on rotated digits	confirmed (0.86–0.88 vs 0.50–0.55)